

Nominal Capacity (Ah)	Metrics	CapCLR (Ours)	MLP	LSTM	TCN	CNN	iTransformer	TimesNet	DLinear	PatchTST	SparseTSF	TimeMixer
15.5	RMSE	0.219±0.007	0.284±0.012	0.297±0.011	0.325±0.015	0.319±0.062	0.235±0.009	0.215±0.003	0.312±0.019	0.427±0.199	0.266±0.009	0.423±0.195
	MAE	0.158±0.003	0.216±0.017	0.222±0.018	0.241±0.016	0.248±0.064	0.174±0.004	0.153±0.004	0.229±0.011	0.358±0.213	0.200±0.006	0.292±0.112
	MAPE	1.07%±0.02%	1.45%±0.11%	1.50%±0.12%	1.62%±0.09%	1.66%±0.41%	1.17%±0.03%	1.03%±0.03%	1.54%±0.07%	2.40%±1.42%	1.34%±0.05%	1.97%±0.75%
20	RMSE	0.776±0.022	1.100±0.109	1.272±0.014	1.292±0.112	1.213±0.019	0.940±0.013	0.855±0.039	1.334±0.015	1.418±0.084	1.052±0.025	0.955±0.194
	MAE	0.560±0.011	0.828±0.124	0.965±0.024	0.960±0.082	0.904±0.015	0.683±0.014	0.629±0.024	1.017±0.038	1.075±0.078	0.784±0.040	0.698±0.165
	MAPE	3.19%±0.07%	4.74%±0.78%	5.48%±0.19%	5.31%±0.36%	5.09%±0.11%	3.91%±0.08%	3.62%±0.17%	5.79%±0.25%	5.84%±0.39%	4.47%±0.33%	4.00%±0.95%
23	RMSE	1.074±0.028	1.277±0.034	1.837±0.058	1.798±0.033	1.519±0.042	1.200±0.006	1.096±0.036	1.877±0.058	1.590±0.098	1.319±0.040	1.131±0.024
	MAE	0.786±0.012	0.988±0.033	1.428±0.056	1.378±0.033	1.176±0.031	0.904±0.014	0.811±0.028	1.486±0.064	1.239±0.096	1.018±0.034	0.844±0.004
	MAPE	4.55%±0.07%	5.65%±0.14%	8.14%±0.33%	7.89%±0.15%	6.62%±0.12%	5.19%±0.08%	4.69%±0.21%	8.48%±0.33%	6.78%±0.46%	5.78%±0.19%	4.85%±0.06%
30	RMSE	0.550±0.017	0.843±0.050	1.161±0.294	1.777±0.067	1.168±0.076	0.699±0.079	0.567±0.042	1.192±0.278	1.069±0.525	1.077±0.429	6.341±9.676
	MAE	0.329±0.020	0.551±0.042	0.772±0.213	1.449±0.047	0.856±0.068	0.441±0.049	0.323±0.011	0.704±0.032	0.827±0.578	0.534±0.052	0.853±0.687
	MAPE	1.18%±0.07%	1.97%±0.16%	2.80%±0.77%	5.16%±0.17%	3.07%±0.24%	1.59%±0.18%	1.17%±0.04%	2.55%±0.12%	2.90%±1.91%	1.91%±0.16%	3.15%±2.62%
52	RMSE	1.062±0.044	1.603±0.173	1.695±0.081	2.132±0.116	1.726±0.067	1.268±0.036	1.119±0.019	1.821±0.022	2.616±0.745	1.399±0.064	1.469±0.279
	MAE	0.605±0.028	1.075±0.252	1.038±0.065	1.440±0.168	1.103±0.075	0.769±0.036	0.625±0.027	1.119±0.058	2.273±0.830	0.874±0.104	0.903±0.194
	MAPE	1.23%±0.06%	2.16%±0.48%	2.12%±0.12%	2.94%±0.29%	2.24%±0.15%	1.56%±0.08%	1.26%±0.05%	2.28%±0.12%	4.48%±1.58%	1.78%±0.19%	1.84%±0.40%
63	RMSE	0.994±0.063	1.482±0.081	2.242±0.135	3.040±0.577	2.186±0.116	1.194±0.062	1.171±0.056	2.255±0.244	3.578±0.439	1.340±0.123	1.622±0.361
	MAE	0.477±0.019	0.999±0.039	1.321±0.060	2.679±0.647	1.436±0.181	0.598±0.034	0.606±0.088	1.404±0.108	3.392±0.434	0.751±0.083	0.958±0.311
	MAPE	0.78%±0.03%	1.63%±0.05%	2.19%±0.10%	4.29%±1.01%	2.37%±0.28%	0.98%±0.04%	0.99%±0.13%	2.32%±0.19%	5.37%±0.68%	1.23%±0.13%	1.57%±0.50%
67	RMSE	0.864±0.024	1.283±0.067	1.290±0.082	1.953±0.380	1.465±0.143	1.024±0.214	0.848±0.018	1.452±0.033	3.269±1.944	1.217±0.113	1.367±0.172
	MAE	0.555±0.047	0.892±0.050	0.915±0.060	1.606±0.450	1.107±0.161	0.673±0.199	0.517±0.017	1.063±0.019	3.003±2.030	0.852±0.144	1.009±0.081
	MAPE	0.85%±0.06%	1.37%±0.08%	1.39%±0.09%	2.43%±0.66%	1.68%±0.23%	1.03%±0.30%	0.79%±0.03%	1.62%±0.02%	4.51%±3.03%	1.30%±0.21%	1.54%±0.14%
105	RMSE	2.220±0.045	3.169±0.201	5.298±0.242	6.308±0.065	5.395±0.332	2.329±0.130	2.205±0.071	6.073±0.060	6.255±1.076	3.474±0.156	2.706±0.454
	MAE	1.428±0.033	2.274±0.251	3.648±0.227	4.569±0.155	3.829±0.199	1.536±0.121	1.415±0.098	4.624±0.080	4.941±0.909	2.529±0.165	1.907±0.369
	MAPE	1.78%±0.04%	2.79%±0.28%	4.36%±0.23%	5.35%±0.14%	4.60%±0.25%	1.90%±0.15%	1.76%±0.12%	5.49%±0.07%	5.55%±0.98%	3.09%±0.18%	2.32%±0.41%
Avg.	MAPE	1.83%	2.72%	3.50%	4.37%	3.42%	2.17%	1.91%	3.76%	4.73%	2.61%	2.66%